

Experiment 7:

Write program using the NLTK library to perform part-of-speech tagging on a text.

Program:

```
import nltk

nltk.download('punkt')

nltk.download('punkt_tab')

nltk.download('averaged_perceptron_tagger')

nltk.download('averaged_perceptron_tagger_eng')

text = input("Enter a sentence: ")

words = nltk.word_tokenize(text)

tags = nltk.pos_tag(words)

print("\nPOS Tags:")

for word, tag in tags:

    print(word, ":", tag)
```

Output:

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
Enter a sentence: i love watching movies

POS Tags:
i : NN
love : VBP
watching : VBG
movies : NNS
```

